EXOPLANET HABITABILITY CLASSIFICATION: FIND-S AND CANDIDATE ELIMINATION CASE STUDY

ASSIGNMENT 01

Github link: [nidhi-shree/AML-assignments]

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1. Introduction & Background

The search for habitable exoplanets is one of the most significant scientific endeavors of the 21st century. NASA's Kepler Space Telescope and the TESS mission have catalogued over 5,500 confirmed exoplanets, of which only a fraction fall within the classical habitable zone of their host star. Machine learning concept-learning algorithms, specifically Find-S and Candidate Elimination offer a transparent, rule-based approach to classifying whether an exoplanet may support liquid water and, by extension, life.

This case study applies both algorithms to a curated dataset of 10 real-world-inspired exoplanets drawn from the NASA Exoplanet Archive and the Habitable Exoplanet Catalogue (HEC) maintained by the Planetary Habitability Laboratory, University of Puerto Rico. The target concept is: Is this exoplanet Potentially Habitable? (Yes / No).

1.1 Why This Domain?

- Purely categorical attributes make the hypothesis space tractable for concept learning.
- The habitable zone concept maps naturally to conjunctive attribute constraints.
- Real datasets (Kepler-442b, Proxima Centauri b, TRAPPIST-1e) provide grounded examples.
- Failure cases arise naturally from conflicting stellar and planetary conditions.

1.2 Key Terminology

Term	Definition
Instance Space (X)	All possible exoplanet descriptions across 6 categorical attributes.
Hypothesis Space (H)	All conjunctive attribute-value constraints over X.
Version Space (VS)	Subset of H consistent with all labelled training examples.
S boundary	Most specific hypotheses consistent with positive examples.
G boundary	Most general hypotheses are consistent with all examples.
Habitable Zone (HZ)	The orbital range where liquid water can exist on a planet's surface.
ESI	Earth Similarity Index – composite score of planetary similarity to Earth.

2. Exoplanet Dataset

2.1 Attribute Schema

Six categorical attributes are used, each discretized from real-valued measurements available in the NASA Exoplanet Archive. This discretization mirrors the approach used by the Habitable Exoplanet Catalogue's ESI scoring tiers.

Attribute	Values	Meaning
StarType	G-Type | K-Type | M-Type	Spectral class of host star (Sun-like → Red Dwarf)
ZonePosition	Inner | Habitable | Outer	Orbital position relative to the stellar habitable zone
PlanetSize	Sub-Earth | Earth-Like | Super-Earth | Giant	Radius relative to Earth
Atmosphere	Thick | Thin | None	Modelled atmospheric density
SurfaceWater	Likely | Possible | Unlikely	Probability of liquid water based on temperature models
MagneticField	Strong | Weak | None	Estimated magnetosphere protection from stellar radiation

2.2 Training Examples

The 10 training examples below are inspired by real exoplanets. Positive examples (Yes) are confirmed, or strongly candidate habitable worlds; negative examples (No) are confirmed hostile environments.

#	Planet	StarType	Zone	Size	Atmos.	SurfWater	MagField	Label
D1	Kepler-442b	K-Type	Habitable	Earth-Like	Thin	Likely	Strong	Yes
D2	Kepler-452b	G-Type	Habitable	Super-Earth	Thick	Likely	Strong	Yes
D3	55 Cancri e	G-Type	Inner	Super-Earth	Thick	Unlikely	Weak	No
D4	TRAPPIST-1e	M-Type	Habitable	Earth-Like	Thin	Likely	Strong	Yes
D5	HAT-P-7b	F-Type*	Inner	Giant	Thick	Unlikely	None	No
D6	Kepler-186f	M-Type	Habitable	Earth-Like	Thin	Possible	Strong	Yes
D7	WASP-12b	G-Type	Inner	Giant	None	Unlikely	None	No
D8	Proxima Centauri b	M-Type	Habitable	Earth-Like	Thin	Possible	Weak	Yes
D9	GJ 1132b	M-Type	Inner	Sub-Earth	None	Unlikely	None	No
D10	TOI-700d	M-Type	Habitable	Earth-Like	Thin	Likely	Strong	Yes

** HAT-P-7b's host star is an F-type (hotter than the Sun); treated as a distinct category for this study.*

3. Find-S Algorithm: Exoplanet Habitability

3.1 Concept

Find-S begins with the most specific hypothesis h = <∅,∅,∅,∅,∅,∅>, matching no planet and generalizes it each time a positive (habitable) example is encountered. Negative examples (hostile planets) are ignored. The output is the single most specific hypothesis consistent with all confirmed habitable worlds in the training set.

3.2 Step-by-Step Trace

Step / Planet	Hypothesis h	Reasoning
Initial	<∅, ∅, ∅, ∅, ∅, ∅>	Start — matches nothing
After D1 (+) Kepler-442b	<K-Type, Habitable, Earth-Like, Thin, Likely, Strong>	Initialise to D1 values
After D2 (+) Kepler-452b	<?, Habitable, ?, Thin, Likely, Strong>	StarType: K≠G → ? | Size: Earth-Like≠Super-Earth → ?
After D3 (−) 55 Cancri e	<?, Habitable, ?, Thin, Likely, Strong>	Negative — ignored
After D4 (+) TRAPPIST-1e	<?, Habitable, ?, Thin, Likely, Strong>	All match → no change
After D5 (−) HAT-P-7b	<?, Habitable, ?, Thin, Likely, Strong>	Negative — ignored
After D6 (+) Kepler-186f	<?, Habitable, ?, Thin, ?, Strong>	SurfaceWater: Likely≠Possible → ?
After D7 (−) WASP-12b	<?, Habitable, ?, Thin, ?, Strong>	Negative — ignored
After D8 (+) Proxima b	<?, Habitable, ?, Thin, ?, ?>	MagneticField: Strong≠Weak → ?
After D9 (−) GJ 1132b	<?, Habitable, ?, Thin, ?, ?>	Negative — ignored
After D10 (+) TOI-700d	<?, Habitable, ?, Thin, ?, ?>	All match → no change

Find-S Final Hypothesis:

```
h = <?, Habitable, ?, Thin, ?, ?>

Interpretation: A planet is potentially habitable if and only if
it orbits within the Habitable Zone AND has a Thin atmosphere.
All other attributes are unconstrained by positive examples alone.
```

3.3 Interpretation

The result aligns well with current planetary science: orbital position within the habitable zone is the primary discriminator, and a thin (Earth-like) atmosphere is necessary for surface liquid water. Find-S correctly identifies these two features but cannot use negative examples to further constrain the hypothesis.

3.4 Limitations in This Domain

- Ignores hostile planets entirely: cannot rule out giants or magma worlds.

- Misses the importance of the magnetic field which is crucial for shielding stellar radiation from M-type stars.
- Produces a single hypothesis; cannot quantify uncertainty across the version space.

4. Candidate Elimination Algorithm: Exoplanet Habitability

4.1 Concept

Candidate Elimination maintains two boundary sets that bracket the version space: S (most specific consistent hypotheses) and G (most general consistent hypotheses). Unlike Find-S, it uses both positive and negative examples. Positive examples generalize S; negative examples specialize G. The algorithm converges when S = G, yielding a single learnable concept.

4.2 Initialization

S_0 = { <∅, ∅, ∅, ∅, ∅, ∅> } (most specific matches nothing)
G_0 = { <?, ?, ?, ?, ?, ?> } (most general matches everything)

4.3 Full Trace

Step / Planet	S boundary	G boundary
Initial	{ <∅,∅,∅,∅,∅,∅> }	{ <?,?,?,?,?,?> }
D1 + Kepler-442b <K-Type,Habitable,Earth-Like,Thin,Likely,Strong>	{ <K-Type,Habitable,Earth-Like,Thin,Likely,Strong> }	{ <?,?,?,?,?,?> }
D2 + Kepler-452b <G-Type,Habitable,Super-Earth,Thick,Likely,Strong>	{ <?,Habitable,?,?,Likely,Strong> }	{ <?,?,?,?,?,?> }
D3 − 55 Cancri e <G-Type,Inner,Super-Earth,Thick,Unlikely,Weak>	{ <?,Habitable,?,?,Likely,Strong> }	{ <K-Type,?,?,?,?,?>, <M-Type,?,?,?,?,?>, <?,Habitable,?,?,?,?>, <?,?,Earth-Like,?,?,?>, <?,?,?,Thin,?,?>, <?,?,?,?,Likely,?>, <?,?,?,?,?,Strong> }
D4 + TRAPPIST-1e <M-Type,Habitable,Earth-Like,Thin,Likely,Strong>	{ <?,Habitable,?,?,Likely,Strong> }	{ <?,Habitable,?,?,?,?>, <?,?,?,?,Likely,?>, <?,?,?,?,?,Strong> }
D5 − HAT-P-7b <F-Type,Inner,Giant,Thick,Unlikely,None>	{ <?,Habitable,?,?,Likely,Strong> }	{ <?,Habitable,?,?,?,?>, <?,?,?,?,Likely,?>, <?,?,?,?,?,Strong> }
D6 + Kepler-186f <M-Type,Habitable,Earth-Like,Thin,Possible,Strong>	{ <?,Habitable,?,?,?,Strong> }	{ <?,Habitable,?,?,?,?>, <?,?,?,?,?,Strong> }
D7 − WASP-12b <G-Type,Inner,Giant,None,Unlikely,None>	{ <?,Habitable,?,?,?,Strong> }	{ <?,Habitable,?,?,?,?>, <?,?,?,?,?,Strong> }
D8 + Proxima b <M-Type,Habitable,Earth-Like,Thin,Possible,Weak>	{ <?,Habitable,?,?,?,?> }	{ <?,Habitable,?,?,?,?> }
D9 − GJ 1132b <M-Type,Inner,Sub-Earth,None,Unlikely,None>	{ <?,Habitable,?,?,?,?> }	{ <?,Habitable,?,?,?,?> }
D10 + TOI-700d <M-Type,Habitable,Earth-Like,Thin,Likely,Strong>	{ <?,Habitable,?,?,?,?> }	{ <?,Habitable,?,?,?,?> }

Converged Version Space (S = G):

```
S = G = { <?, Habitable, ?, ?, ?, ?> }

Learned Concept Rule:
  A planet is POTENTIALLY HABITABLE if and only if
  its orbital position is within the HABITABLE ZONE.

All other attributes remain unconstrained — consistent with
the known diversity of candidate habitable worlds (e.g., varying
star types from Kepler-452b's G-type to TRAPPIST-1e's M-type).
```

4.4 Scientific Interpretation

The converged rule ZonePosition = Habitable is the single strongest discriminator in our dataset, matching the core definition of habitability in planetary science. The algorithm correctly identifies that star type, planet size, atmosphere, and magnetic field are too variable across confirmed habitable candidates to be fixed constraints with the given 10 examples. This mirrors real scientific uncertainty: Proxima b (weak magnetic field, M-type star) and Kepler-452b (G-type, thicker atmosphere) are both candidate habitable worlds despite differing substantially.

5. Five Situations Where Version Space Cannot Be Obtained

In real-world exoplanet classification, the version space is impossible or meaningless to obtain in the following five situations. Each is grounded in actual challenges faced by the Planetary Habitability Laboratory and NASA's exoplanet research.

5.1 Conflicting Labels from Observational Uncertainty

Exoplanet measurements carry significant error margins. Two planets with identical discretized attribute values may receive conflicting habitability labels depending on which observational model is used (e.g., optimistic vs. conservative habitable zone definitions). This introduces contradictory training examples that collapse the version space to empty.

```
Real Example:
 D_a: <M-Type, Habitable, Earth-Like, Thin, Possible, Weak> → YES
    (optimistic HZ definition — Proxima Centauri b scenario)
 D_b: <M-Type, Habitable, Earth-Like, Thin, Possible, Weak> → NO
    (conservative HZ definition — same planet, tidal locking concern)

 No hypothesis in H can simultaneously satisfy both labels.
 S cannot generalise past D_a; G cannot specialise to exclude D_b.
 → Version Space = ∅ (empty)
```

5.2 Disjunctive Target Concept (Multiple Habitable Zone Types)

The classical habitable zone assumes liquid water as the solvent. However, scientists now consider subsurface oceans (e.g., Europa-like moons) and exotic chemistries. The true habitability concept may be a disjunction: (ZonePosition=Habitable) OR (Interior=IceCovered AND SubsurfaceOcean=Yes). Conjunctive hypothesis spaces cannot represent disjunctions.

```
Real Example:
  Habitable planets split into two disjoint groups:
    Group A: Surface water worlds (ZonePosition=Habitable)
    Group B: Subsurface ocean worlds (ZonePosition=Outer, IcyCrust=Yes)

  No single conjunctive hypothesis covers both groups.
  Find-S would over-generalise to <?,?,?,?,?,?> (everything).
  CE would find S = G = <?,?,?,?,?,?> — meaningless.
  → True concept lies outside H; VS = entire H (useless)
```

5.3 Only Positive Examples Available (No Confirmed Hostile Planets)

Early in a mission (e.g., the first year of Kepler data), astronomers may have confirmed habitable-zone planets but no confirmed hostile ones. Without negative examples, the G boundary never specialises below <?,?,?,?,?,?>, making the version space infinitely broad.

```
Real Example:
  Training set: D1, D2, D4, D6, D8, D10 — all labeled YES.

  After processing:
```

```
    S = { <?, Habitable, ?, Thin, ?, ?> }  (generalised from positives)
    G = { <?, ?, ?, ?, ?, ?> }             (never constrained)

  Every hypothesis between S and G is 'consistent' — including ones
  that classify gas giants as habitable.
  → Version Space = entire H between S and G (not informative)
```

5.4 Only Negative Examples Available (No Confirmed Habitable Planet)

Conversely, if all confirmed exoplanets in a catalogue are hostile (e.g., hot Jupiters, lava worlds), S remains at the null hypothesis <∅,∅,∅,∅,∅,∅>. It never rises because Find-S has nothing to initialise from, and CE's S boundary has no positive example to generalise.

```
Real Example:
 Training set: D3, D5, D7, D9 — all labeled NO.

 After processing:
  S = { <∅,∅,∅,∅,∅,∅> }  (never moved — no positive example)
  G shrinks but S stays null → version space has no lower bound.

 Any query planet would be predicted 'Unknown' since S matches nothing.
 → Version Space = ∅ (from below) — algorithm produces no learnable
concept
```

5.5 Continuous / High-Dimensional Real-Valued Attributes

The actual NASA Exoplanet Archive stores attributes as continuous real values: orbital period in days (0.4 – 700+), planet radius in Earth radii (0.3 – 25+), equilibrium temperature in Kelvin (100 – 3000+), stellar luminosity, etc. The hypothesis space over continuous attributes is infinite and cannot be enumerated. S and G boundary sets become infinite, making exact CE impossible.

```
Real Example:
 Attributes: OrbitalPeriod (days), PlanetRadius (R⊕), EqTemperature (K),
        StellarLuminosity (L⊙), StellarMass (M⊙), SurfaceGravity (g)

 Hypothesis space size:
  For just OrbitalPeriod: infinitely many threshold pairs [a, b]
  For 6 continuous attrs: S and G each have ∞ members.

 CE cannot maintain exact boundaries → algorithm fails.
 Modern approaches (SVMs, Decision Trees) handle this instead.
 → Version Space = mathematically defined but computationally
unobtainable
```

6. Code Reference

The accompanying Python script bank_loan_candidate_elimination.py — renamed exoplanet_candidate_elimination.py — implements both algorithms on the exoplanet dataset. Key functions:

Function	Description
`find_s(training_data)`	Initialises h=<∅…∅>, generalises on each positive example.
`candidate_elimination(training_data)`	Maintains S/G boundaries; updates on positive & negative examples.
`matches(h, instance)`	Returns True if hypothesis h covers the given planet instance.
`min_generalisations(h, inst)`	Returns minimal generalisations of h to cover positive planet inst.
`min_specialisations(g, inst)`	Returns minimal specialisations of g to exclude negative planet inst.
`classify(instance, S, G)`	Classifies a new planet using the converged version space.
`demo_failure_cases()`	Demonstrates all 5 VS failure scenarios with exoplanet examples.

7. Conclusion

Applied to exoplanet habitability classification, both algorithms converge on the same core finding: orbital position within the habitable zone is the single strongest conjunctive discriminator learnable from 10 training examples. Find-S additionally identifies a thin atmosphere as a constraint (from positive examples alone), while Candidate Elimination relaxes this after accounting for negative examples, producing a more generalizable rule.

The five failure cases highlight real challenges in astrobiology data science: observational uncertainty causes contradictory labels; disjunctive habitability concepts (surface vs. subsurface oceans) exceed conjunctive hypothesis spaces; early missions lack negative examples; and real NASA data is continuous rather than categorical. These limitations motivate the use of more expressive models, Random Forests, SVMs, and neural networks for production exoplanet classification.

References

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